

Single Source Justification (Simplified Acquisition)

Section A: General Contract Information

Is this a Bridge Action as defined in the [AF Bridge Action Reduction Plan](#)?

☐ Yes

☒ No

Contracting Activity: AFRL/PZLET

Project/Program Name: Material State Awareness Laboratory Ultrasound Modeling Capability Enhancement

Purchase Request / Local ID Number: TBD

Section B: Description of the Supplies/Services Required

The Materials and Manufacturing Directorate, Materials State Awareness Branch (AFRL/RXNW) has a requirement for the purchase of one (1) Commercial License for Pogo Software. Pogo is a general elastodynamic finite element solver to model linear acoustic waves and vibrations interacting with physical structures. Pogo incorporates pre- and post-processing tools to generate input models and view the outputs.

In order to avoid unnecessary delays, the requested software license is the only item available that will satisfy the Government's requirement at this time. This action will be:

Contract Type: FFP

Funding: TBD

Period of Performance: Base Year - 12 Months, Option Year I - 12 Months

Contract Number: TBD

This is a sole source requirement with Pogo Software identifying the following company as its exclusive distributor:

I C Consultants Ltd (Imperial Consultants)
58 Princes Gate
Exhibition Road
London
SW7 2PG, United Kingdom

Items to be acquired include:

Description	Qty	
Pogo Software License (Base year)	1	
Pogo Software License (Option Year)	1	

Section C: Justification for Soliciting from a Single Source or Brand Name

Inspection of composite bondlines is key for current and future Department of the Air Force (DAF) aerospace vehicles. These inspections are often done using ultrasonic testing (UT), and the complexity of the materials systems make it challenging to design them without the aid of high performance simulations. There are many commercial finite element analysis (FEA) software packages that can perform UT simulations. Examples include [REDACTED]. These software packages are primarily designed to perform computations on the CPU, which works well for "static" computational problems (e.g. stress analysis) that can leverage implicit solvers. However, UT simulations are dynamic, as the practitioner is modeling the propagation of an ultrasonic wave over time. This requires a time-domain explicit solver, which is inherently more computationally expensive. This challenge is exacerbated by the frequency of the UT inspection being simulated, as the frequency of typical DAF inspections is on the order of high kHz to MHz. This results in very fine discretization (in time and space) of the model, which increases the computational cost. Ultimately, traditional commercial FEA solvers are not designed to

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efficiently perform UT simulations. In fact, the AFRL/RXNW team has in the past had to deploy simulations using traditional solvers to the high performance computing center (HPC) to simulate inspection scenarios.

A recent alternative is available through Pogo. Pogo is a commercial FEA solver that is designed for UT simulations. The key difference between Pogo and other packages is that Pogo is designed to perform computations on a GPU. GPUs are designed for substantial parallel processing, making a GPU-based solver for UT simulations significantly more efficient than CPU-based solvers. With Pogo, the RXNW team can perform complex 3D simulations on local computing nodes and deliver results in hours vs. weeks (or without having to access limited HPC resources).

Accordingly, Pogo is the only software capable of providing UT simulations described above without the U.S. Air Force experiencing significant mission impacts and delays, as well as substantial duplication of costs that could not be recouped through competition.

Section D: Efforts to Obtain Competition

An intent to award sole source (IASS) notice was posted on SAM.gov [REDACTED]. [REDACTED]
[REDACTED]

No additional market research was conducted due to Pogo Software providing a sole source, original equipment manufacturer (OEM) letter identifying Imperial Consultants as the exclusive distributor of its software products to the US Federal Government.

This letter is located in File Element #340 of the contract file.

Section E: Steps to Preclude Future Single Source or Brand Name Awards

To the best of the government's ability, the leading competing products available in the current market place have been evaluated prior to purchasing the solution. AFRL/RXNW will continue to monitor the market to determine if other products meet the Government's needs. AFRL/RXNW will continue to conduct market research using catalogues, the internet, attend industry events, trade journal searches, market surveys and other resources to locate other solutions to meet the government's requirements.

Section F: Contracting Officer's Determination

I have determined, in accordance with [FAR 13.106-1\(b\)\(1\)](#), that the circumstances of this contract action deem only a single source is reasonably available.

[REDACTED]	[REDACTED]	[REDACTED]
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